1. Installation of NVIDIA CUDA Toolkit:-

- sudo apt update
- sudo apt install nvidia-cuda-toolkit
- nvcc –version

2. Check NVIDIA GPU

- lspci | grep -i nvidia

3. Installation of GNU Compiler Collection (GCC)

- gcc –version
- sudo apt install gcc

4. Installation of Linux Headers

- sudo apt-get install linux-headers-$(uname -r)

5. Installation of NVIDIA Driver

- sudo add-apt-repository ppa:graphics-drivers/ppa --yes
- sudo apt update
- sudo apt install nvidia-driver-470
- sudo apt install nvidia-driver-495 (for 12.2 version)
- `nvidia-smi`

```
root@ip-172-31-85-10:~# nvidia-smi
Tue Jun  4 09:45:26 2024
+---------------------------------------------------------------------------------------+
| NVIDIA-SMI 535.171.04             Driver Version: 535.171.04   CUDA Version: 12.2     |
|-----------------------------------------+----------------------+----------------------+
| GPU  Name                 Persistence-M | Bus-Id        Disp.A | Volatile Uncorr. ECC |
| Fan  Temp   Perf          Pwr:Usage/Cap |         Memory-Usage | GPU-Util  Compute M. |
|                                         |                      |               MIG M. |
|=========================================+======================+======================|
|   0  NVIDIA A10G                    Off | 00000000:00:1E.0 Off |                    0 |
|  0%   36C    P0             62W / 300W |  14286MiB / 23028MiB |      0%      Default |
|                                         |                      |                  N/A |
+-----------------------------------------+----------------------+----------------------+

+---------------------------------------------------------------------------------------+
| Processes:                                                                            |
|  GPU   GI   CI        PID   Type   Process name                            GPU Memory |
|        ID   ID                                                             Usage      |
|=======================================================================================|
|    0   N/A  N/A     40246      C   python3                                  14278MiB |
+---------------------------------------------------------------------------------------+
```

AI 1.3 Server Code Path:- /home/ubuntu/service/stable-diffusion/sd-gen-service

```
root@ip-172-31-85-10:/home/ubuntu/service/stable-diffusion/sd-gen-service# ls -la
total 60
drwxr-xr-x 6 ubuntu ubuntu 4096 Apr 11 10:56 .
drwxr-xr-x 4 ubuntu ubuntu 4096 Aug 11  2023 ..
drwxr-xr-x 7 ubuntu ubuntu 4096 Aug 14  2023 .git
-rw-r--r-- 1 ubuntu ubuntu   67 Aug 14  2023 .gitignore
drwxrwxr-x 6 ubuntu ubuntu 4096 Aug 14  2023 .venv
-rw-r--r-- 1 ubuntu ubuntu 1037 Jun  4 11:08 config.ini
-rw-rw-r-- 1 ubuntu ubuntu 5548 Feb  3 23:15 index.py
-rw-r--r-- 1 ubuntu ubuntu  894 Aug 14  2023 install.sh
drwxr-xr-x 9 ubuntu ubuntu 4096 Jan  6 06:03 model
drwxr-xr-x 4 ubuntu ubuntu 4096 Feb  2 00:34 modules
lrwxrwxrwx 1 ubuntu ubuntu   52 Aug 11  2023 output -> /home/ubuntu/service/stable-diffusion/aws-s3/output/
-rw-r--r-- 1 ubuntu ubuntu 3489 Aug 14  2023 readme.md
-rw-r--r-- 1 ubuntu ubuntu   83 Aug 14  2023 requirements.txt
-rw-r--r-- 1 ubuntu ubuntu   84 Aug 14  2023 requirements_cuda.txt
-rw-rw-r-- 1 ubuntu ubuntu  484 Aug 14  2023 sd-gen.service
lrwxrwxrwx 1 ubuntu ubuntu   54 Aug 16  2023 transfer -> /home/ubuntu/service/stable-diffusion/aws-s3/transfer/
```

S3fs Mounted Bucket Path :- /home/ubuntu/service/stable-diffusion/aws-s3

- Then we have to create the symbolic links between the aws-s3 (/home/ubuntu/service/stable-diffusion/aws-s3) path to (/home/ubuntu/service/stable-diffusion/sd-gen-service)

```
root@ip-172-31-85-10:/home/ubuntu/service/stable-diffusion/aws-s3# ll
total 8
drwxrwxr-x 4 ubuntu ubuntu 4096 Aug 16  2023 ./
drwxr-xr-x 4 ubuntu ubuntu 4096 Aug 11  2023 ../
drwxrwxrwx 1 ubuntu ubuntu    0 May 31 19:51 output/
drwxrwxrwx 1 ubuntu ubuntu    0 May 31 17:34 transfer/
```

Ai-transfer bucket is mounted on transfer bucket and output bucket is mounted on output bucket

Command for mounted the Transfer Bucket in AI –1.3: -

s3fs ai-transfer /home/ubuntu/service/stable-diffusion/aws-s3/transfer -o passwd_file=/home/ubuntu/.passwd-s3fs

Command for mounted the Output File Bucket in AI – 1.3: -

s3fs ai-output-bucket /home/ubuntu/service/stable-diffusion/aws-s3/output -o passwd_file=/home/ubuntu/. passwd-s3fs

Command for mounted the Transfer Bucket in AI –1.2: -

s3fs ai-input-bucket /home/ubuntu/service/stable-diffusion/aws-s3/output-file -o passwd_file=/home/ubuntu/. passwd-s3fs

Command for mounted the Output File Bucket in AI – 1.2: -

s3fs ai-transfer /home/ubuntu/service/stable-diffusion/aws-s3/output-file -o passwd_file=/home/ubuntu/.passwd-s3fs

We can verify to hit command df –h (for AI-1.3)

s3fs 4.0G 0 4.0G 0% /home/ubuntu/service/stable-diffusion/aws-s3/transfer
s3fs 4.0G 0 4.0G 0% /home/ubuntu/service/stable-diffusion/aws-s3/output

```
root@ip-172-31-85-10:/home/ubuntu/service/stable-diffusion/aws-s3# df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root         97G   54G   44G  55% /
devtmpfs         7.8G     0  7.8G   0% /dev
tmpfs            7.8G  4.0K  7.8G   1% /dev/shm
tmpfs            1.6G  1.2M  1.6G   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
tmpfs            7.8G     0  7.8G   0% /sys/fs/cgroup
/dev/loop2        56M   56M     0 100% /snap/core18/2812
/dev/loop1        26M   26M     0 100% /snap/amazon-ssm-agent/7993
/dev/loop0        26M   26M     0 100% /snap/amazon-ssm-agent/7983
/dev/loop6        92M   92M     0 100% /snap/lxd/24061
/dev/loop4        64M   64M     0 100% /snap/core20/2264
/dev/loop7        39M   39M     0 100% /snap/snapd/21759
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
/dev/loop3        56M   56M     0 100% /snap/core18/2823
/dev/loop8        39M   39M     0 100% /snap/snapd/21465
/dev/loop5        64M   64M     0 100% /snap/core20/2318
tmpfs            1.6G  4.0K  1.6G   1% /run/user/1000
tmpfs            1.6G  4.0K  1.6G   1% /run/user/1001
tmpfs            1.6G   24K  1.6G   1% /run/user/124
tmpfs            1.6G     0  1.6G   0% /run/user/0
s3fs             4.0G     0  4.0G   0% /home/ubuntu/service/stable-diffusion/aws-s3/transfer
s3fs             4.0G     0  4.0G   0% /home/ubuntu/service/stable-diffusion/aws-s3/output
```

We can verify to hit command df –h (for AI-1.2)

s3fs 4.0G 0 4.0G 0% /home/ubuntu/service/stable-diffusion/aws-s3/transfer-file
s3fs 4.0G 0 4.0G 0% /home/ubuntu/service/stable-diffusion/aws-s3/output-file

```
root@ip-172-31-84-115:~# df -h
Filesystem       Size  Used Avail Use% Mounted on
/dev/root         97G   65G   33G  67% /
devtmpfs         7.8G     0  7.8G   0% /dev
tmpfs            7.8G  8.0K  7.8G   1% /dev/shm
tmpfs            1.6G  984K  1.6G   1% /run
tmpfs            5.0M     0  5.0M   0% /run/lock
tmpfs            7.8G     0  7.8G   0% /sys/fs/cgroup
/dev/loop0        25M   25M     0 100% /snap/amazon-ssm-agent/7628
/dev/loop2        56M   56M     0 100% /snap/core18/2823
/dev/loop1        26M   26M     0 100% /snap/amazon-ssm-agent/7993
/dev/loop3        56M   56M     0 100% /snap/core18/2829
/dev/loop4        64M   64M     0 100% /snap/core20/2182
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
/dev/loop5        92M   92M     0 100% /snap/lxd/24061
/dev/loop6        41M   41M     0 100% /snap/snapd/20671
/dev/loop7        64M   64M     0 100% /snap/core20/2318
/dev/loop8        39M   39M     0 100% /snap/snapd/21759
tmpfs            1.6G  4.0K  1.6G   1% /run/user/1001
tmpfs            1.6G  4.0K  1.6G   1% /run/user/1000
tmpfs            1.6G     0  1.6G   0% /run/user/0
s3fs             4.0G     0  4.0G   0% /home/ubuntu/service/stable-diffusion/aws-s3/transfer-file
s3fs             4.0G     0  4.0G   0% /home/ubuntu/service/stable-diffusion/aws-s3/output-file
```

Creating the virtual environment to run the index.py file

Source .venv/bin/activate
Python3 index.py

```
root@ip-172-31-85-10:/home/ubuntu/service/stable-diffusion/aws-s3# tmux ls
AI: 1 windows (created Mon Jun  3 19:03:59 2024)
```